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Integrated Technologies to Improve Low-Permeability Coalbed Methane Production Outcomes and Coal Mine Methane Pre-Drainage Efficiency and Emissions Reductions

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Abstract

This paper presents a comprehensive evaluation of emerging technologies designed to enhance coalbed methane (CBM) production, improve coal mine gas pre-drainage for safety, and reduce emissions of coal mine methane (CMM). The integrated framework and workflows combine advanced diagnostic, drilling, and fracturing techniques to address the challenges of developing low-permeability coal seams. Broader applications of Diagnostic Fracture Injection Tests (DFITs) are explored, extending their use beyond conventional stress and reservoir characterisation to include the identification of Pressure-Dependent Leakoff (PDL) and directional permeability through multi-well interference testing. This enables improved stress profiling, drainage pattern assessment, and more informed development planning. In addition, the paper discusses the evolution of indirect hydraulic fracturing and tight radius drilling (TRD) to enhance mine gas pre-drainage and emissions control, supported by innovations in micro-proppant materials and placement strategies to maintain fracture conductivity in coal seams. Field results demonstrate how DFITs and indirect fracturing improved pre-drainage at an underground Bowen Basin mine. Complementing this, the paper reviews the historical use and future potential of Tight Radius Drilling (TRD) and Surface-to-In-Seam (SIS) wells combined with reservoir barriers to further lower CMM emissions. This integrated approach brings together proven, tested, and modelled technologies suited to the unique conditions of the Surat and Bowen Basins and other low-permeability, highly stressed coal reservoirs in the Eastern Hemisphere. The novel contribution lies in presenting these methods as a cohesive workflow for CBM and CMM operators seeking practical, accessible solutions to unlock challenging reserves while enhancing safety and reducing greenhouse gas emissions. Barrier implementation models are also highlighted, indicating the potential for open-cut mines to achieve up to 46% emissions reduction per square kilometre of drainage area, demonstrating substantial environmental benefits alongside production gains.

Background

Australia has advanced many technologies, enabling the most developed Coalbed Methane (CBM) industry in the world and improvements in Coal Mine Methane (CMM) management and pre-drainage. Cross-pollination of researchers and practitioners across the CBM and mining industries has been a key element in advancing many of these technology improvements, as well as the readiness to adopt best practices developed from each industry. CBM has enabled Australia to be a major Liquefied Natural Gas (LNG) exporter, and CMM drainage has transformed problematic gas into a valuable mine site resource. Beneficial use of gas at the mine site has improved profitability and demonstrates the industry's social license to operate in an increasing 'coal-unfriendly' world. Finally, as pressure increases internationally to reduce methane emissions, Australian technologies are being developed that can aid the mining industry to meet increasingly stringent emissions reduction targets.

To illustrate the workflow and benefits possible from applying emerging technologies, one has first to understand the past and the contributing factors leading to the need for a change or improvement. In Australian coals, this was precipitated by significant advances in understanding of: (1) the coal fabric and its impact on gas storage and transport (Dawson & Esterle, 2010; Hamilton et al., 2012; Scott, 2002); (2) the stress and structural state (Enever et al., 1994; Enever & Lee, 2000; Hillis et al., 1999; Mukherjee et al., 2020); then (3) coupling that data with an understanding of the natural cleat and fracture network and its interactions with the current stress state and impacts on development (Johnson et al., 2002; Kirk-Burnnand et al., 2015; Tavener et al., 2017). In this paper, the authors propose that there is a systematic methodology that can be employed in CBM/CMM pilot wells to understand key reservoir uncertainties and guide prudent activities to progress pilot activities towards a development decision effectively.

Low-permeability coal seam gas (CSG) wells typically require stimulation to reach commercial production, with project success largely dependent on managing pressure-dependent permeability (PDP) behaviour, fracture effectiveness, and stimulated reservoir volume (SRV). These factors are closely linked and require integrated workflows for effective evaluation and optimisation. PDP behaviour, common in early production, must be characterised using Diagnostic Fracture Injection Tests (DFITs), reservoir simulation, and detailed coal property analysis, with larger DFITs providing deeper insights into pressure-dependent leak-off. Fracture effectiveness is critical for sustaining productivity, requiring the maintenance of fracture length and conductivity while minimising complexity to avoid issues like proppant embedment and fines generation. Enhancing SRV and fracture quality through informed design and technology application is key to overcoming the inherent challenges of low-permeability coals.

An Improved Workflow for CBM/CMM Well/Pilot Planning

The acquisition and processing of the data for both CBM and CMM well planning is a multi-disciplined process involving geoscience, petroleum and reservoir engineering skills. Several technologies derived from coal seam gas (CSG), and the petroleum industry can be adapted to improve mine pre-drainage, improve mine safety and generate gas for beneficial mine use or third-party sales. These techniques include: (1) understanding the geology and stratigraphy; (2) developing an understanding of stress state and reservoir parameters using a diagnostic fracture injection test (DFIT), (3) an expanded Stimulated Reservoir Volume (SRV) using multi-stage, horizontal well, hydraulic fracturing techniques, (4) TRD implementation for areas where hydraulic fracturing is inefficient, or SIS wells limited by compartmentalisation; and (5) micro-proppant technology in conjunction with hydraulic fracturing or standalone treatments. The following paragraphs and [Figure 1](#) outline the integrated technology framework required to provide the best opportunity to manage the challenges in developing low-permeability coals.

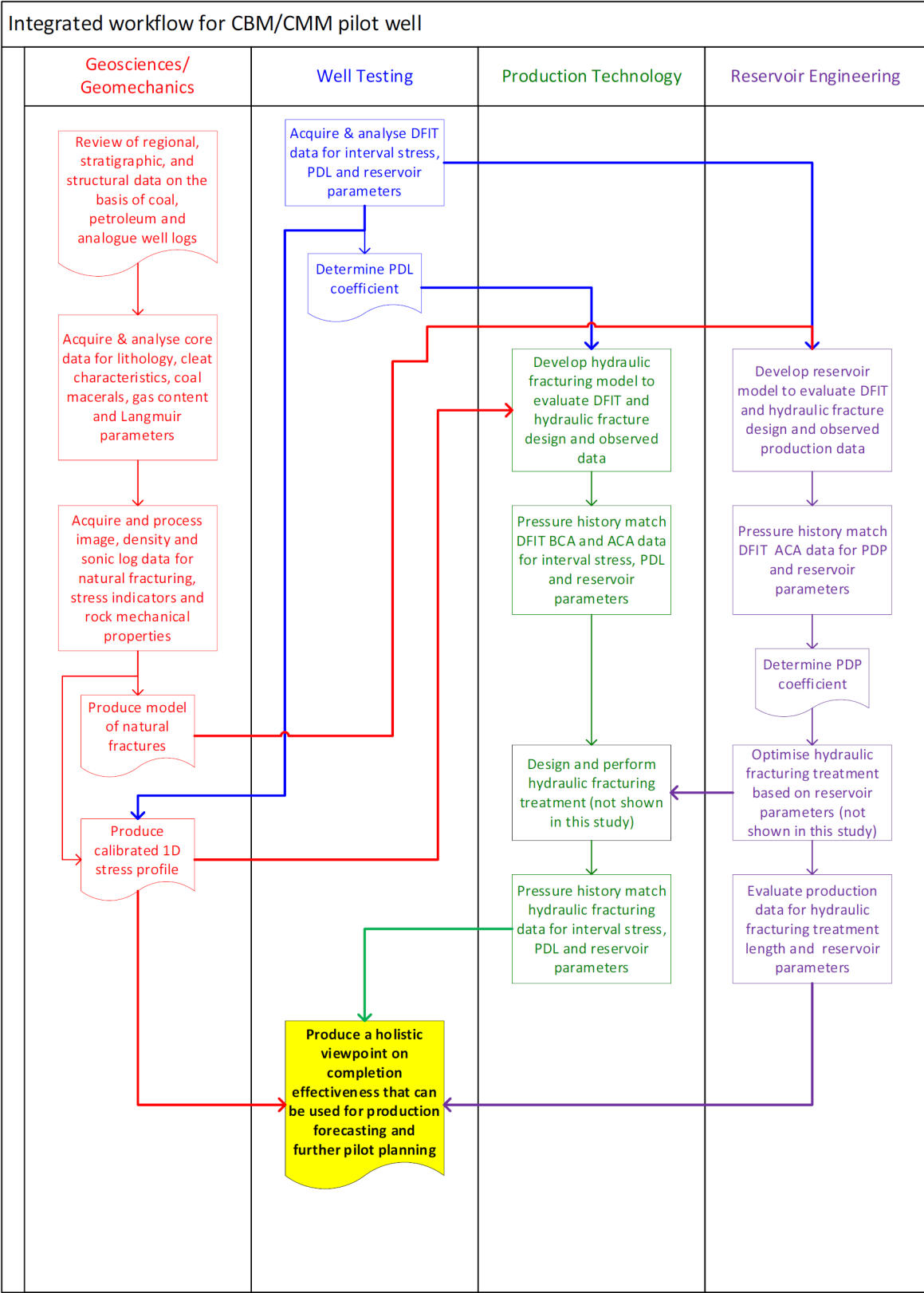


Figure 1—Integrated workflow for CBM/CMM pilot well with key elements by discipline.

Importance of Geologic/Petrology Data

The geological setting of the basin of interest has to be studied to understand the tectonic history, the deformation style, the burial history, and faulting that shaped the coal sequences (Allen & Allen, 2005).

Detailed structural mapping is necessary to understand possible compartmentalisation of the fields and the impact on mine planning and well placement. Fault characterisation is crucial to predict flow or non-flow behaviour and the deformational properties during possibly necessary stimulations and later production (e.g. Undershultz et al., 2019, Murray & Power, 2021, Zhou et al., 2022). This is the first stage of evaluation in the integrated workflow for CBM/CMM pilot well planning and implementation (see Fig. 1).

The next stage in the geological evaluation is acquisition of coal and characterizing the coal properties by proximate and petrographic analyses (e.g. O'Brian et al., 2003, Gupta, 2007, Brosk et al, 2020). Core analysis should include lithology description, maceral analysis, cleat characterisation (Laubach et al., 1998), gas content measurements, and determination of the Langmuir parameters. Coal rank, thickness distribution of the seams and continuity will trigger exploitation activities. This is the second step in the first swim lane in Figure 1.

During the drilling, data must be collected to correlate seams and confirm their continuity and thickness, which informs field delineation for CBM targeting. Geophysical well log analysis will help to understand the distribution of coal seams and interlayered sandstones, which might exhibit permeable gas saturated zones (e.g. Kennedy, 2015; Tiab & Donaldson, 2016; Jin et al., 2022). A logging program should include density, neutron porosity, sonic and resistivity data to perform the necessary petrophysical analysis. Density and sonic data are needed to determine rock mechanical properties.

Finally, the last step of the geological evaluation is the acquisition and interpretation of micro-resistivity image log data, which allows the identification of natural fractures/cleat systems and stress indicators, which feed into the reservoir and geomechanical model (Laubach et al., 1998; Zhou et al., 2022), respectively. These models produce a framework, from which, when coupled with diagnostic fracture injection test (DFIT) data, can be used to develop a calibrated 1D stress profile and understand stress sensitivity, necessary inputs to the design of hydraulic fracture treatments, and constrain reservoir models to evaluate fracture effectiveness. A fully coupled stress and reservoir model, along with the geological and structural findings, defines target identification and the choice of the most effective well orientation and completion.

Importance of Diagnostic Fracture Injection Test (DFIT) Data

DFIT data is essential for accurately profiling stress as well as characterising pressure-dependent permeability (PDP) in low-permeability reservoirs. Advanced workflows have been developed to integrate DFIT data to improve stress profiling as well as reservoir modelling (Johnson Jr. et al. 2020, Johnson and Saldungaray 2025). The process of determining the key parameters is illustrated in the second vertical swim lane in Figure 1, labelled "Well Testing".

DFITs provide critical information across different pressure regimes: before fracture closure, they assess permeability and derive the pressure-dependent leak-off (PDL) constant based on exponential leak-off to natural fractures; after closure, they estimate initial permeability through Darcy flow, aiding in reservoir parameter calibration. These insights allow for more accurate quantification of permeability loss under high stress and enable modelling of potential improvements through coal shrinkage effects, making DFITs a cornerstone in optimising stimulation and production strategies. T

A workflow developed by Johnson Jr., You, et al. (2020) offers a comprehensive, multidisciplinary approach to coal reservoir characterisation, aiming to improve the accuracy of post-fracturing performance forecasts through tightly constrained history matching. This process integrates static and dynamic data, starting with geological descriptions of cleat fabric and porosity from image logs and core analysis, followed by the development of permeability anisotropy and stress-dependent changes using regional or analogue data. DFITs are central to the workflow, providing data to build stress profiles, derive pressure-dependent leak-off (PDL) parameters, and calibrate permeability loss under net effective stress. Treating pressure history-matching and fracture dimensions are determined by modelling the DFIT using a 3D fracture model that incorporates PDL and PDP effects; this process is noted by the first two boxes in the third swim lane in Figure 1, labelled "Production Technology".

Thereafter, using the natural fracture model and reservoir characterisation data, the after-closure DFIT data are also used to constrain fracture compressibility in a pressure-dependent reservoir simulator; this process is noted by the first three boxes in the final swim lane in [Figure 1](#), labelled "Reservoir Engineering". Ultimately, these steps improve the hydraulic fracture treatment design and thereafter validate the reservoir model by history-matching production data. Integrating a DFIT into multi-well interference testing offers a powerful method for characterising coal seam permeability, anisotropy, and formation connectivity. The DFIT-based pulse is robust enough to history-match using numerical simulation to effectively evaluate interference behaviour ([Johnson Jr. et al. 2025](#)). By using DFIT-derived fracture dimensions as the pulse source in a pressure-dependent reservoir simulator (PDRS), the pressure response in offset observation wells can be analysed to inform optimal well spacing and enhance field development planning.

Hydraulic Fracturing and Stimulation Advances

Fracture complexity is a common problem in coal stimulations and is related to several geological and geomechanical factors and often becomes manifest in the fracture initiation by high NWBPL. Firstly, multimode hydraulic fracture generation (Sneddon 1946, Atkinson and Thiercelin 1993) is highly dependent on the degree of natural fracturing/cleating, the geomechanical framework and overarching in situ stress regimes, and is frequently evidenced in Australian coal hydraulic fracturing studies ([Leonardi et al. 2019](#), [Johnson, Glassborow, et al. 2010](#), [Johnson et al. 2002](#), [Busetti and Flottman 2018](#), [Morales and Davidson 1993](#), [Morales et al. 1990](#)). Australian coals have also demonstrated complex far-field geometries related to interactions with boundaries, discontinuities, and jointing ([Jeffrey et al. 1992](#), [Jeffrey et al. 1995](#), [Jeffrey et al. 1998](#), [Johnson, Scott, et al. 2010](#), [Flottman et al. 2013](#)). These studies highlight the importance of understanding the geological framework and coupling that with observations from the DFIT to develop an accurate picture of the degree of complexity that might be observed in the hydraulic fracturing treatment, as outlined in the previous paragraphs.

To implement changes or improvements in design strategies for low-permeability coals, some improvements have been made in vertical fracturing technologies. Previous studies have identified that 'best-practices' include: (1) maintaining good standoff from coal boundaries to reduce chances of complexity ([Cramer 1992](#)); minimize the number of intervals and perforated sections to improve fracture distribution in multi-seam sequences ([Johnson, Scott, et al. 2010](#), [Scott et al. 2010](#)); understanding of the overall stress regime, interlayer stress contrasts and potential complexities arising from differing current and paleo stress environments ([Johnson, Glassborow, et al. 2010](#), [Johnson 1995](#), [Flottman et al. 2013](#), [Johnson et al. 2002](#)); and reducing complexity by optimising the well azimuth relative to the current stress environment ([Bentley et al. 2013](#)). Two emerging technologies to increase the stimulated reservoir volume for low-permeability coals for both CBM and CMM applications have been the use of Indirect Hydraulic Fracturing (IHF) techniques ([Johnson Jr. and Ramsay 2024](#), [Johnson et al. 2021](#), [Ramanandraibe et al. 2023](#)) and the implementation of micro-proppants ([Johnson and Saldungaray 2025](#), [Johnson Jr 2024](#), [Ramanandraibe et al. 2022](#), [Johnson et al. 2022](#)).

Indirect Hydraulic Fracturing (IHF) Process. Fracture complexity in coal seams is influenced by natural cleating, geological structures, and in situ stress regimes, often revealed through high net wellbore pressure loss (NWBPL) during fracture initiation. Australian hydraulic fracturing studies frequently show multimode fracture generation and complex far-field geometries due to interactions with boundaries, discontinuities, and jointing. These complexities challenge conventional fracture treatments, particularly in high-stress coal environments. Indirect hydraulic fracturing (IHF) was originally proposed for simplifying fracture geometry in North American coals ([Olsen et al. 2007](#)) but has shown greater effectiveness in compressional or transpressional regimes, where fractures can migrate from adjacent formations into lower-stress coal seams. Recent applications of multi-stage IHF from horizontal wells have emerged due to challenges with wellbore instability in coals and the inability to use traditional completion methods in mining zones. To

date, 54 IHF stages have been placed across vertical and horizontal wells in the Cooper and Bowen Basins, with successful initiation and low NWBPL but limited proppant placement in coal seams.

The first IHF well in Australia, the Jaws 1ST well, was drilled in the Cooper Basin as a sidetrack for an abandoned in-seam drilling attempt at the same location (Branajaya et al. 2019), caused by coal seam instability and loss of drilling bottom hole assembly. This IHF well was drilled ~3m below the seam and showed improved performance compared to other wells with seven (7) IHF frac stages coupled with aggressive heel and toe electric submersible pump artificial lift. While all stages were successfully placed, one stage was never producing or recovering load fluid, based on liquid tracing, as a result of overdisplacement.

For CMM applications, the interconnecting of Surface-to-In-seam (SIS) wells has been shown to be more effective in CMM drainage and could benefit underperforming SIS wells in CBM applications. Their application to date has been deployed between parallel SIS wells to drain potential roadways and panels adequately to drive roadways. Two important cases of indirect hydraulic fracturing techniques highlighted the benefits of using IHF on a mine pre-drainage basis to interconnect and better pre-drain SIS wells along planned gate roads to allow gas contents low enough to drive the roadways (Johnson Jr. and Ramsay 2024, Johnson et al. 2021). In both of these applications, the IHF well was targeted to place the steel casing and frac hardware > 2 metres below the coal seam floor, to reduce the potential intersection during mining operations. A side benefit noted by Johnson Jr & Ramsey (2024) was the interconnection and pre-drainage of an underlying seam that should also reduce floor gas during longwall mining operations.

Ramanandraibe et al. (2022) studied the optimisation of IHF and potential SIS intercept well spacing. They saw increased estimated ultimate recoveries (EURs) with increasing well spacing and a similar fracture half-length interconnection between the IHF well and two offsetting SIS wells. Notwithstanding, Net Present values (NPVs), using estimated current costs and gas prices, increased dramatically with the number of frac stages, with up to twenty (20) stages in a one (1) km IHF lateral. The data for spacings >100m and the time of production required to drain those areas effectively are more applicable for CBM applications rather than CMM pre-drainage spacings. In CMM cases, well spacings are in the order of 50-60 metres between parallel SIS wells, which are intersected and used to drain potential roadways and panels adequately.

IHF Well Planning and Execution. Johnson et al. (2021) outlined the preparatory work and execution steps for the initial indirect hydraulic fracturing (IHF) treatment, which built upon earlier vertical well IHF trials, mined-back evaluations, and extensive data acquisition, including permeability and stress testing.

Initially, a vertical well should be drilled to obtain site-specific DFITs and drill stem test (DST) data to derive localised PDL parameters, stress magnitudes, and stress directions. Post-DFIT imaging logs, along with breakdown and closure pressure data, support the construction of a localised 1D stress profile through best-fit solutions across multiple strata using Kirsch, poroelastic, and fissure pressure models. Following testing, the vertical well should be completed open-hole through the coal and interburden, with optional casing (PVC or GRE) cemented in unstable or sensitive formations for isolation.

Key planning considerations for IHF and SIS well integration included aligning wellbores orthogonal to maximum horizontal stress to promote transverse fracture development, with optimal inter-well spacing based on vertical well microseismic or surface deformation tiltmeter data. The SIS well is typically drilled first to assess seam continuity and gather LWD data, such as azimuthal gamma ray and compressional sonic logs. As aforementioned, it is recommended that the IHF well be placed within 2 metres above or below the seam and guided by roof/floor touches if azimuthal gamma ray data are unavailable. Once the vertical well is intersected, and a frac liner with sleeves spaced 40–50 m apart is installed, guided by treatment modelling and fracture design objectives.

Thereafter, the IHF well should be drilled using a similar approach to the to-be-intercepted SIS wells, with behind-the-bit gamma-ray steering and floor touches for navigation. Sleeve spacing in the completion should be optimised to minimise back stress from previous stages. Fracture design considerations included

appropriate proppant selection and concentration to ensure fractures are adequately propped, reduce embedment, limit fines generation, and incorporate breaker technologies to preserve fracture conductivity. A properly designed IHF and intersecting SIS well are pictured in Figure 2.

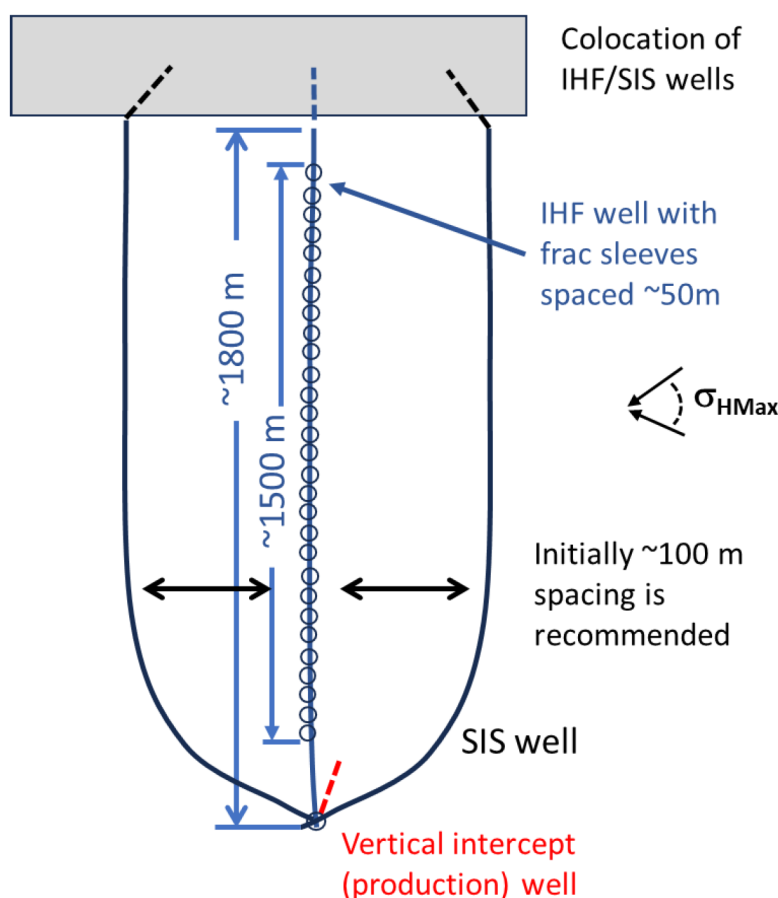


Figure 2—IHF/SIS design strategy (plan view) using two intersecting SIS wells (Johnson Jr, 2024).

History of Micro Proppants in CBM/CMM Applications. PDL is indicative of fracture dilation and enhanced permeability in coal as a result of low net effective stress. Conventional-sized proppants used in complex coal fracture environments, including 100 mesh sand, are unable to prop natural fractures and cleats effectively. These unpropped fractures typically close during flowback and increasing net effective stress, as shown in earlier studies (Johnson, Glassborow, et al. 2010). While 100 mesh sand can reduce fluid leak-off, laboratory (Penny and Conway 1995) and modelling studies—including Lattice-Boltzmann simulations (Di Vaira et al. 2022)—confirm minimal proppant invasion into natural fractures. In particular, particles larger than 325 mesh (50 μm) have repeatedly shown no meaningful penetration or propping effect in coals under lab leak-off conditions (Keshavarz et al. 2014, Johnson et al. 2022).

To enhance recovery in low-permeability coals, hydraulic fracturing can be coupled with graded particles or commercially available micro-proppants. Research indicates that 10–20 μm particles can maintain or even improve cleat conductivity under (Keshavarz et al. 2015, Khanna et al. 2013). More recent tests using 10–50 μm micro-proppants in both Permian coal and synthetic coal analogues demonstrated reduced fluid loss and preserved conductivity in stress-sensitive fractures (Johnson et al. 2022). However, further laboratory evaluation is necessary to define the effective particle size range fully. Until then, particles >325 mesh should not be classified as functional micro-proppants for coal reservoir stimulation.

Determining the Benefits and Field Application of Micro Proppants. A CBM or CMM reservoir with characteristics indicating pressure-dependent permeability (PDP) is likely to be a key indicator

for potentially successful micro-proppant application. Studies, as mentioned earlier, have shown that conventional fracturing technologies in high PDL environments with fracture complexity have failed to deliver a stimulated reservoir volume adequate to achieve commercial gas flows (Johnson, Scott, et al. 2010, Johnson, Glassborow, et al. 2010, Johnson et al. 2002, Flottmann et al. 2018, Flottman et al. 2013). Further studies have documented the benefits of micro-proppants from standalone, vertical fracturing and horizontal well applications (Johnson Jr et al. 2022, Johnson, Leonardi, You, Ribiero, et al. 2022, Santiago et al. 2021, You, Wang, Di Vaira, et al. 2019, You, Wang, Leonardi, et al. 2019, Ribeiro et al. 2019).

Two studies in particular (Santiago et al. 2022, Ramanandraibe et al. 2022) linked the PDL volume and the benefits directly to the impacted region around the vertical fracture as a result of total fluid leak off in the invasion region. The Ramanandraibe et al. (2023) study indicated the enhanced permeability region around the fracture and resulting benefit in NPV were only slightly improved with the use of micro proppants in the case of an indirect IHF and SIS well connections. However, anecdotal evidence of indirect connectivity between IHF and SIS wells, like those observed by Johnson Jr. & Ramsey (2024), would not likely remain connected without micro proppants being present in that fluid phase.

Finally, Johnson et al (2022) highlight the operational logistics and planning that will be required to deploy commercially available micro proppant products in the field. Since that report, two frac treatments using commercially available materials were pumped. They showed anecdotal evidence of increased cleanup post-frac (Johnson and Saldungaray 2025), relative to previous deep coal stimulations in the Cooper Basin, Australia, at shallower depths (Camac et al. 2018). Recent changes in the commercially available material, employing particle agglomeration, will provide more confidence in the process of field managing these extremely fine-meshed materials and seamlessly deploying the technology with little additional dust issues relative to conventional proppants. Again, the reader is encouraged to review the publicly available NERA research report that provides comprehensive workflows and also guidance on the design, execute, and evaluation of micro proppant applications (Johnson, et al. 2022).

Post-Processing Modelling and Optimisation Steps

The workflow steps for post-processing and modelling integrate data acquisition, drilling, and completion activities to inform field-wide development strategies. They are noted by interchanges in the flow of the last two swim lanes of Figure 1. As previously mentioned, DFITs are critical to develop realist models in a pressure-dependent reservoir simulator (PDRS) that incorporates the DFIT-derived PDP parameters to constrain and realistically develop the range of deliverability estimates. Probabilistic approaches can allocate hydraulic fracture benefits based on vertical well simulations and, if applicable, IHF-SIS well interactions. Initial modelling may use average fracture conductivity to assess production and spacing outcomes. Iterative calibration between fracture and reservoir simulators ensures alignment of static properties, stress magnitudes, and dynamic reservoir characteristics, helping to refine pilot well designs and fracture placement strategies.

Real-time data from SIS and monitoring wells around the fracture treatments further refine the fracture models and effective stimulated reservoir area estimates. These models, once calibrated post-treatment, are reintegrated into the reservoir simulator for history matching and sensitivity analysis, enabling informed development planning. For CMM pre-drainage optimisation, the process, as demonstrated by Ramanandraibe et al. (2023), supports maximising EUR by balancing drainage area and permeability while optimising relative to mining schedules. Overall, the workflow outlined in Figure 1 culminates in a flexible yet robust model that drives further optimisation of subsequent pilot programs and scales to field-wide deployment in a field development planning framework.

CMM Pre-Drainage and Emissions Reduction Strategies

Australia has advanced many technologies in the areas of Coal Mine Methane (CMM) management and pre-drainage and has the most developed Coalbed Methane (CBM) industry in the world. Many of the technology improvements in both CMM and CBM production have been the result of cross-pollination of researchers and practitioners across the mining and CBM area, as well as the readiness to adopt best practices developed from each industry. CMM drainage and CBM production over the last twenty years have transformed problematic gas into a valuable mine site resource and Liquefied Natural Gas (LNG) as a major Australian export.

The CRC Mining group at The University of Queensland (UQ) and, more recently, Novus have been developing the Tight Radius Drilling (TRD) technology to complement SIS in improving pre-drainage (Adam and Kok 2015, Meyer et al. 2004) and to reduce emissions from shallow coals in surface mining applications (Johnson Jr. and Mason 2025). Further, UQ and Novus have defined the application and potential benefits of employing reservoir barriers to lower CMM emissions for surface and underground mining applications (Alarcon Vigil et al. 2025, Johnson Jr and Mirzaghobanali 2024, Johnson Jr 2023, Johnson Jr. 2022, Johnson Jr 2014). The combination of these and other petroleum engineering skills and technologies is enhancing safety by more effective pre-drainage and reducing greenhouse gas emissions.

SIS and TRD Technology

CRC Mining developed TRD as an alternative completion technology to increase production and improve recovery using a horizontally deployable jetting tool from a vertical wellbore, thereby increasing the contact area between the coal and the production wellbore. Individual radially jetted channels can extend over 250 meters. They can be deployed over multiple seams from a single borehole (see Fig. 3). Multiple laterals are created radially by rotating the deployment head, and their combined length can exceed 1500 meters, offering a competitive edge compared to SIS wells in a single seam. Additionally, TRD allows for jetting across multiple coal seams from a single vertical well, reducing costs compared to drilling multiple SIS wells.

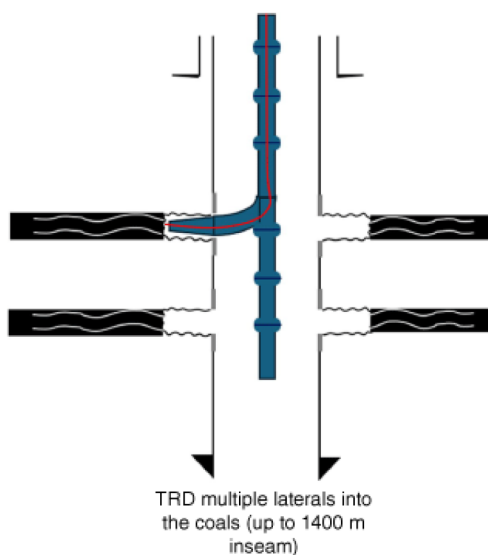


Figure 3—Pre-drainage techniques include tight radius drilling deployable in multiple seams (Johnson and Mason, 2025)

TRD was proven effective in placing multiple laterals to pre-drain coals at the Anglo Grasstree and BMC South Walker Creek (SWC) mine sites (Adam & Kok, 2015; Meyer et al., 2004). At Anglo's Capcoal Mining Complex, TRD was used to pre-drain a longwall and between the two vertical shafts at the time of the expansion into the Grasstree Mine from the adjoining Southern Colliery. At the time of application, gas

contents were at undrained levels based on an igneous dyke separating Grasstree and the Southern Colliery. Meyer et al. (2014) noted that GT#1 drilled 16 laterals with a total length of 1165 metres, with seven (7) laterals over 100 metres long (see Figure 5) and 1300 metres, with the longest being 182 metres long. An author of this paper personally observed several of the TRD lateral intersections in the Grasstree Mine, exhibiting widths greater than the 64mm tool diameter and similar to that pictured by Meyer et al. (2014) of 150mm diameter.

At the BMC SWC five-well trial deployment, 12 to 17 laterals per well of 700 to 1754 total lateral lengths per well, respectively (see Figure 3), with the average and furthest lengths being in excess of 90m and 300m, respectively, achieving continuous drilling rates of over 240 m/hr, and a drilling rate per shift increasing from 70 to 280 m/shift throughout the campaign (Adam & Kok, 2015). Following the drilling, the wells were placed in production and produced over 12,000 m³ of water and 150,000 m³ of gas over a roughly 3-year period. Discussions with the former CRC Mining engineer indicated that BMA indicated that the wells could have produced better with more aggressive production management (Adam, 2024). Recently, data from this production test were history-matched in a desorption-capable, reservoir model implementing a representative model of an average TRD pattern that indicated the technology effectively drained a 100m × 400m area of coal on a per-average well basis, respectively (see Fig. 4).

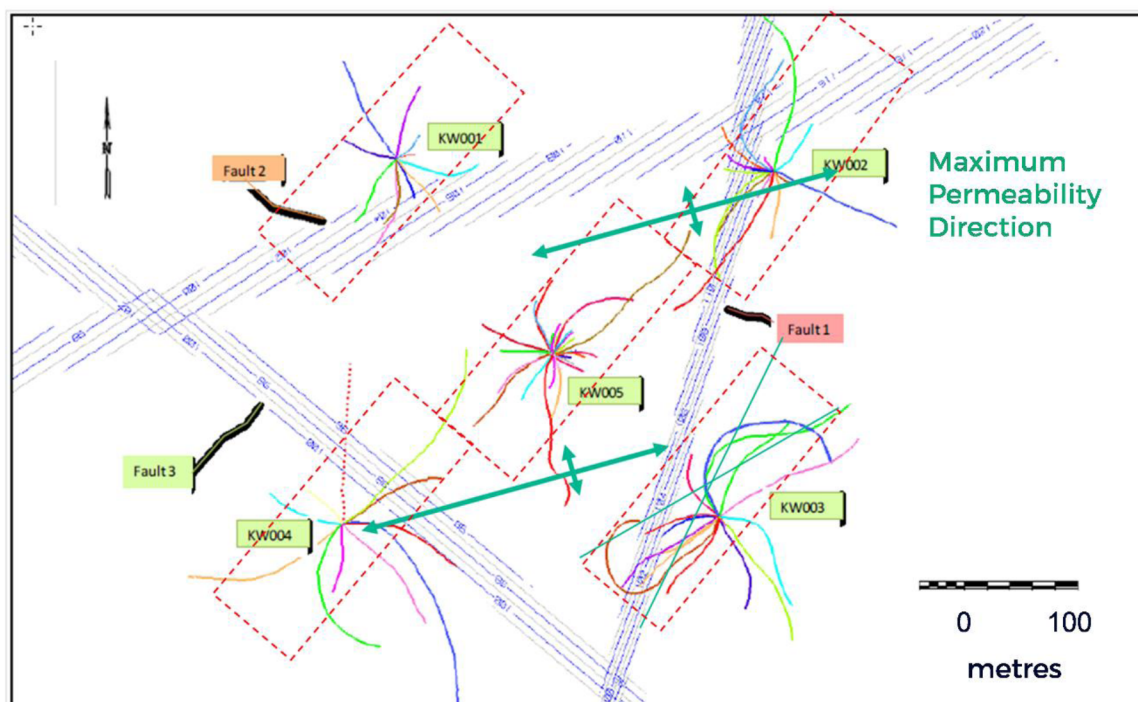


Figure 4—Plan view of five well SWC trials with history-matched drainage areas (red dotted boxes) and maximum permeability direction as inferred from multi-well interference testing (After Adam and Kok 2015)

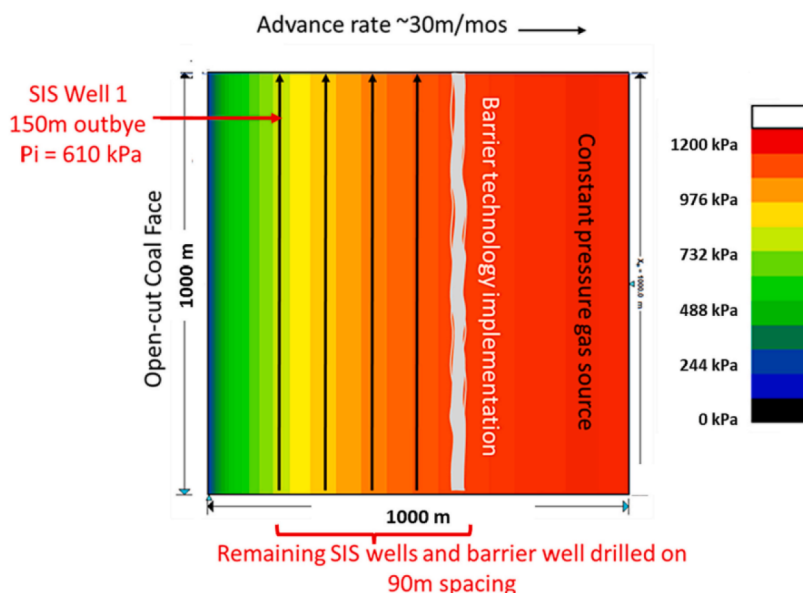


Figure 5—SIS well pattern with barrier application to prevent higher pressure, down-dip gas influx (Johnson, 2024)

Barrier Implementation and Emissions Modelling

Modelling for both underground and surface applications confirm that implementing subterranean barrier wells with SIS drainage reduces fugitive emissions, controls unwanted gas inflows, and preserves gas for future production when it is more economic to recover or for ongoing beneficial use. For underground mining, modelling confirms that a properly placed barrier can achieve a high blockage efficiency (up to 99.9%) by drastically reducing coal seam permeability in the barrier zone (Johnson Jr, 2014), preventing unwanted gas influx into mine works. This maintains higher pressures behind the barrier, improving drainage efficiency and retaining more gas for planned extraction. The key factor for effectiveness is coal permeability—high-permeability seams to benefit most. The barrier does not significantly affect gas drainage efficiency related to the seam dip, but the dip can impact water production rates, which affects drainage costs and gas economics.

Recent reservoir modelling studies of a 1 square kilometre open-cut mine section show that SIS wells alone drain about 24% of gas in the near-face zone but allow additional gas to migrate in from down-dip areas (see Figure 4). This uncontrolled influx temporarily adds about 7% more production from any SIS drainage wells but ultimately results in 69% of the gas being lost during mining (Johnson Jr and Mirzaghobanali, 2024; Johnson Jr, 2023; R. L. Johnson, 2023). By adding a barrier well, the same 24% is recovered for beneficial use near the mining face. This modelling predicts that 57% of the remaining gas is available for later staged recovery; this cuts immediate emissions losses to just 19%. This approach preserves gas for future extraction in better coordination with mining progress, improving net recovery and can enhance project economics (Alarcon Vigil et al., 2025). Ultimately, the gas in barrier blocks further down-dip is prevented from migrating to the face, thereby reducing emissions as shown in Figure 5.

CMM Beneficial Use

Beneficial use of coal mine methane (CMM) at mine sites has significantly improved operational profitability and strengthened the industry's social license to operate in an increasingly carbon-conscious world. As global pressure mounts to reduce methane emissions, Australian technologies, such as Tight Radius Drilling (TRD) and subterranean barrier implementations, offer scalable solutions to convert fugitive methane into usable energy. These technologies enable flaring, onsite power generation, and diesel displacement, and are adaptable across diverse geological settings. Their applicability extends beyond Australia to major coal-producing countries including China, India, Russia, and the United States.

Field demonstrations have shown promising results. For example, pilot projects have successfully powered mine haul fleets using captured methane, achieving substantial diesel reductions (Korobchuk et al., 2021). These outcomes highlight the dual benefits of emissions abatement and improved mine economics.

However, to unlock broader international investment, regulatory frameworks must evolve to align incentives with real-time emissions reductions. As discussed in Alarcon Vigil et al. (2025), the timing of carbon credit issuance is critical. Issuing credits at the point of gas extraction or flaring, rather than post-mining benchmarks, can significantly improve project cash flow and investment attractiveness. This is especially relevant as carbon markets expand, ESG scrutiny intensifies, and demand for low-carbon inputs such as "green steel" grows.

Incorporating site-specific emissions estimation methods, as outlined in Alarcon Vigil et al. (2025), can further enhance the credibility and transparency of CMM mitigation projects. These methods support accurate reporting, better planning for beneficial use, and more robust emissions reduction claims under mechanisms like Australia's Safeguard Mechanism.

Ultimately, aligning policy design with technological capabilities will be key to unlocking global methane mitigation opportunities. When combined with effective pre-drainage and barrier strategies, beneficial use of CMM transforms an environmental liability into an economic asset, advancing both sustainability and profitability in the coal mining sector.

Conclusions and Recommendations

Low-permeability CBM wells require stimulation to achieve commercial production rates and determining permeability and the degree of PDP are still major hurdles to optimising completion designs and improving the efficiency of the overall appraisal process. DFITs remain the key source of permeability data. Incorporating PDRS analyses with ACA data provides a better understanding of coal properties and constrains fracture compressibility, a key model parameter required to define PDP behaviour to production. Key factors associated with the success or failure of a low-permeability coal project are often centred around PDP, fracture effectiveness, and overall SRV.

New hydraulic fracturing technologies have been developed to enhance the SRV in coals, using horizontal wells with indirect, multi-stage hydraulic fracturing in excess of 20 stages. These well stimulation technologies can be co-applied in new or existing CBM fields where direct drilling of laterals has or is believed to be risky, and areas where overlapping tenements limit conventional, steel-based completion strategies within the coal. With indirect hydraulic fracturing, further considerations need to be made regarding stress contrast; interlayer effects (i.e., above and below the coal); the fluid sensitivity or formation stability of the 'hosting' layer for the indirect wellbore; and the need to limit over flushing and spacer volumes when placing frac balls and stages. The only known stage failure to date in an IHF treatment was attributed to over-flushing the treatment.

Finally, recent laboratory testing (Johnson, Leonardi, You, Rabeiro, et al. 2022) and modelling of micro-proppants have provided insight into proppant transport, embedment, and screen-out behaviour of micro-proppants. Their use can significantly increase the overall SRV of the system, and studies have outlined key considerations in their application and determining their potential benefits. Additional laboratory work with core and commercially available micro-proppants under hydraulic fracturing (i.e., lower net effective stress conditions) is still desirable to better understand the straining behaviour under such conditions and the expected ranges of improvement away from the main fracture in ancillary fracturing.

Pre-drainage techniques such as SIS, TRD and UIS drilling reduce in-situ CMM concentrations before mining. Hydraulic fracturing and indirect hydraulic fracturing methods further improve gas drainage by creating stimulated reservoir volumes that interconnect existing wells, boosting gas flow from poorly drained zones. Effective emissions management for coal mining combines accurate measurement with

proactive mitigation and beneficial gas use. A thorough site-specific and economic evaluation is essential to turn mitigation efforts into real financial and environmental gains.

Recent studies have shown that incorporating barrier implementations with pre-drainage strategies can reduce emissions for both surface and underground mining applications. Barrier implementations limit uncontrolled gas migration, reduce emissions during mining, and preserve gas for later recovery, improving safety and economics. The barrier performance depends on proper placement, permeability reduction, and real-world verification (e.g., pilot wells, monitoring). The design must account for local geology, actual pressures, reservoir heterogeneity, whilst considering operational costs and well placement. For underground mines, CMM emissions reduction, coal seam permeability is the biggest factor controlling how much a barrier helps high-permeability seams benefit most.

Finally, CMM management strategies, including direct gas sales, onsite power generation, diesel displacement, and flaring, not only reduce greenhouse gas emissions but also unlock positive cash flow opportunities. The optimal mix of flaring, gas sales, power generation, or diesel displacement depends on factors such as the mine's location, available gas reserves, infrastructure, and prevailing market conditions. When combined with effective pre-drainage and barrier methods, these approaches enable mines to transform an environmental liability into an economic advantage, improving both sustainability performance and operational profitability.

References

- Adam, S. and Kok, J. 2015. Application of Continuous Drilling Technologies in Coal Mining. Paper presented at the 2015 Coal Operators' Conference, Woolongong, 11/2/2015. <https://ro.uow.edu.au/cgi/viewcontent.cgi?article=2233&context=coal>.
- Alarcon Vigil, Daniel, Johnson Jr, Raymond L., and Tauchnitz, James. 2025. Improved estimation methods for surface coal mine methane emissions for reporting, beneficial use, and emission reduction purposes and relative to Australia's safeguard mechanisms. *Journal of Environmental Management* **376**: 124366. <https://www.sciencedirect.com/science/article/pii/S0301479725003421>.
- Allen, P.A. & Allen, J.R. (2005): Basin Analysis. Principles and Applications. Second edition. – Blackwell Publishing.
- Bentley, P.J.D., Megorden, M.P., and Jiang, H. 2013. Improving Hydraulic Fracture Geometry by Directional Drilling in Coal Seam Gas Formation, SPE 167053. Paper presented at the SPE Unconventional Resources Conference and Exhibition-Asia Pacific, Brisbane, Australia, 2013/11/11. SPE.
- Branajaya, R., Archer, P., and Farley, A. 2019. Pushing the boundaries – deployment of innovative drilling, completion and production technology to advance a deep coal seam play, Extended Abstract AJ18115. *The APPEA Journal* **59** (2): 770–775. <https://www.publish.csiro.au/paper/AJ18115>.
- Brosk, F.A., Fakhri, H.I., Abdulsattar, M.J. and Hamed, M.J. (2020): The Proximate Analysis method for the Composition Determination of Different Coal Types. - 6th international engineering conference " Sustainable Technology and Development", *IEC-2020 proceedings*, 91–96.
- Camac, B. A., Benson, J., Chan, V. et al. 2018. Cooper Basin Deep Coal - the New Unconventional Paradigm: Deepest Producing Coals in Australia. *ASEG Extended Abstracts* **2018** (1): 1–7. https://doi.org/10.1071/ASEG2018abM3_1A.
- Cramer, D. D. 1992. The Unique Aspects of Fracturing Western US Coalbeds. *SPE-0897-0842-JPT*: 1126–1133.
- Di Vaira, N.J., Łaniewski-Wołk, Ł., Johnson Jr., R.L. et al. 2022. Influence of particle polydispersity on bulk migration and size segregation in channel flows. *Journal of Fluid Mechanics* **939** (A30).
- Flottman, T., Brooke-Barnett, S., Naidu, S.K. et al. 2013. Influence of in Situ Stresses on Fracture Stimulation in the Surat Basin, Southeast Queensland, SPE 167064. Paper presented at the SPE Unconventional Resources Conference and Exhibition-Asia Pacific, Brisbane, Australia, 2013/11/11. SPE.
- Gupta, R. (2007): Advanced Coal Characterization: A Review. – *Energy & Fuels*, 2007, **21**, 451–460.
- Jin, L., Wang, G., Fan, Q., Pang, X., Li, H., Zhao, F., Li, Y., Zhao, X., Zhao, Y., Huang, Y., Bao, M., Qin, Z. and Wang, Q. (2022): Geophysical well-log evaluation in the era of unconventional hydrocarbon resources: a review on current status and prospects. - *Surveys in Geophysics*, **43**, no. 3, 913–957.
- Johnson Jr., R.L. 2022. Reducing coal face methane emissions by implementing subterranean barriers to assist pre-drainage wells. 15/11/2022, The University of Queensland, (Reprint).
- Johnson Jr., R.L., Honari, V., and Alarcon Vigil, D. 2025. Characterisation of Low-Permeability Coal Seam Gas Reservoir Permeability and Permeability Anisotropy using Diagnostic Fracture Injection Testing within a Multi-Well Interference Testing Programme, Paper EP24190. *Australian Energy Producers Journal*.

- Johnson Jr., R.L. and Mason, M. 2025. Tight Radius Drilling in Coal, a Re-Emerging Technology, Paper EP24161. *Australian Energy Producers Journal*.
- Johnson Jr., R.L., You, Z., Ribeiro, A. et al. 2020. Integrating Reservoir Characterisation, Diagnostic Fracture Injection Testing, Hydraulic Fracturing and Post-Frac Well Production Data to Define Pressure Dependent Permeability Behavior in Coal, paper SPE-202330-MS. Proc., SPE Asia Pacific Oil & Gas Conference and Exhibition. <https://doi.org/10.2118/202330-MS>.
- Johnson Jr, R. and Mirzaghorbanali, A. 2024. Improving Pre-Drainage, Reducing Gas Influx, and Lowering Emissions from Surface and Underground Coal Mining Operations Using Subterranean Barrier Implementations. *Proc., International Mine Ventilation Congress 2024*, Sydney, NSW, 12/8/2024, 563–568.
- Johnson Jr, R.L. 2014. Modifying subterranean fluid flow patterns using barriers to improve pre-drainage in coal mining. PhD, Mining Engineering, The University of Queensland.
- Johnson Jr, R.L. 2023. The implementation of subterranean barriers with mine pre-drainage to reduce coal mine methane emissions from open-cut and underground metallurgical-coal mines. Paper presented at the Climate Smart Engineering 2023, Melbourne, VIC, 20/11/2023.
- Johnson Jr, Raymond. 2024. Successful Implementation of Two Decades of Australian Coal Seam Gas Research Can Improve Low-Permeability Coal Productivity and Recovery, Paper urtec-2024-4025119. Paper presented at the SPE/AAPG/SEG Unconventional Resources Technology Conference. <https://doi.org/10.15530/urtec-2024-4025119>.
- Johnson Jr., R.L. and Ramsay, M. 2024. More Effective Pre-Drainage of Low-Permeability Coals Using Indirect Hydraulic Fracturing (IHF) in Co-Application with Surface-to-Inseam (SIS) Wells, Paper 105. Proc., International Mine Ventilation Congress 2024, Sydney, Australia, 12/8/2024.
- Johnson, R., Leonardi, C., You, Z. et al. 2022. Converting tight contingent CSG resources: Application of graded particle injection in CSG stimulation - National Energy Resources Australia (NERA) Project Reports, The University of Queensland, Centre for Natural Gas (4 May 2022).
- Johnson, R.L., Jr. 1995. Fracture Treatment Modifications and Bottomhole Treating Pressure Analysis in the Pictured Cliffs Formation, Rio Arriba County, New Mexico, SPE-29448-MS. Proc., SPE Rocky Mountain Regional/Low-Permeability Reservoirs Symposium, Denver, CO, 20-22 March.
- Johnson, R.L., Jr., Flottman, T., and Campagna, D.J. 2002. Improving Results of Coalbed Methane Development Strategies by Integrating Geomechanics and Hydraulic Fracturing Technologies, SPE 77824-MS. Paper presented at the SPE Asia Pacific Oil and Gas Conference and Exhibition, Melbourne, Australia, 8-10 October. <https://doi.org/10.2118/133066-MS>.
- Johnson, R.L., Jr., Glassborow, B., Datey, A. et al. 2010. Utilizing Current Technologies to Understand Permeability, Stress Azimuths and Magnitudes and their Impact on Hydraulic Fracturing Success in a Coal Seam Gas Reservoir, SPE 133066. Proc., SPE Asia Pacific Oil and Gas Conference and Exhibition, Brisbane, Queensland, Australia, 18-20 October. <http://10.2118/133066-MS>.
- Johnson, R.L., Jr. and Saldungaray, P.M. 2025. The First Application of MPs in Two Deep, Low-Permeability, Permian Coal Stimulations in the Taroom Trough, Bowen Basin, Australia, Paper urtec-2025-4252008. Proc., SPE/AAPG/SEG Unconventional Resources Technology Conference. <https://doi.org/10.15530/urtec-2025-4252008>.
- Johnson, R.L., Jr., Scott, M., Jeffrey, R.G. et al. 2010. Evaluating Hydraulic Fracture Effectiveness in a Coal Seam Gas Reservoir from Surface Tiltmeter and Microseismic Monitoring, SPE 133063. Proc., SPE Annual Technical Conference and Exhibition, Florence, Italy, 19–22 September. <https://doi.org/10.2118/0311-0059-JPT>.
- Johnson, R.L., Ramanandraibe, H.M., Ribeiro, A. et al. 2021. Applications of Indirect Hydraulic Fracturing to Improve Coal Seam Gas Drainage for the Surat and Bowen Basins, Australia, Paper AP-URTEC-2021-208375. Paper presented at the SPE/AAPG/SEG Asia Pacific Unconventional Resources Technology Conference. <https://doi.org/10.15530/AP-URTEC-2021-208375>.
- Kennedy, M. (2015): Practical Petrophysics. – *Developments in Petroleum Science*, v. **62**, 1–416
- Keshavarz, A., Badalyan, A., Carageorgos, T. et al. 2015. Stimulation of coal seam permeability by micro-sized graded proppant placement using selective fluid properties. *Fuel* **144**: 228–236. <https://10.1016/j.fuel.2014.12.054>.
- Keshavarz, A., Yang, Y., Badalyan, A. et al. 2014. Laboratory-based mathematical modelling of graded proppant injection in CBM reservoirs. *International Journal of Coal Geology* **136**: 1–16. <https://10.1016/j.coal.2014.10.005>.
- Khanna, Aditya, Keshavarz, Alireza, Mobbs, Kate et al. 2013. Stimulation of the natural fracture system by graded proppant injection. *J Pet Sci Eng* **111**: 71–77. <http://www.sciencedirect.com/science/article/pii/S092041051300171X>.
- Korobchuk, Daria, Wood, Alex, and Tauchnitz, James. 2021. Improvements in the environmental performance of coal mining vehicle fleets by commercialising incidental waste gas. Paper presented at the CSE 2021: Climate Smart Engineering Conference, Barton, A.C.T. <https://search.informit.com/doi/10.3316/informit.302109328416773>.
- Laubach, S.E., Marrett, R.A., Olson, J.E. and Scott, A.R. (1998): Characteristics and origins of coal cleat: A review. - *Int. J. of Coal Geology*, v. **35**, 175–207.

- Meyer, T., Robertson, B., and Lewis, A. 2004. Tight radius drilling operations at Grasstree Mine. Paper presented at the CRC Mining Conference, Noosa Heads, Queensland, Australia, 5/6/2004. <https://espace.library.uq.edu.au/view/UQ:100041>.
- Mining Education Australia (MEA), Underground Mine Environment Course (2006)
- Murray, T.A. and Power, W.L. (2021): Information Guidelines Explanatory Note: Characterisation and modelling of geological fault zones. - *Report prepared for the Independent Expert Scientific Committee on Coal Seam Gas and Large Coal Mining Development through the Department of Agriculture, Water and the Environment, Commonwealth of Australia*, 2021.
- O'Brian, G., Jenkins, B. Esterle, J. and Beath, H. (2003): Coal characterisation by automated coal petrography. – *Fuel*, **82**, 1067–1073.
- Olsen, T.N., Bratton, T.R., Tanner, K.V. et al. 2007. Application of Indirect Fracturing for efficient Stimulation of Coalbed Methane, SPE-107985-MS. Paper presented at the Rocky Mountain Oil & Gas Technology Symposium, Denver, CO, 2007/1/1/. SPE.
- Penny, G.S. and Conway, M.W. 1995. Coordinated Studies in Support of Hydraulic Fracturing of Coalbed Methane, GRI-95/0283, Gas Research Institute, Chicago, Illinois (September).
- Ramanandraibe, H.M., Johnson, R.L., Sedaghat, M. et al. 2023. Co-Application of Indirect Hydraulic Fracturing and Micro-Proppants with Existing Surface-to-Inseam Wells to Improve Pre-Drainage of Low Permeability Coals in Mining Areas, Paper SPE-216613-MS. Paper presented at the ADIPEC. <https://doi.org/10.2118/216613-MS>.
- Ramanandraibe, H.M., Sedaghat, M., Johnson, R. et al. 2022. Co-application of micro-proppant with horizontal well, multi- stage hydraulic fracturing treatments to improve productivity in the Permian coal measures, Bowen Basin, Australia. *APPEA Journal* **62**: 14.
- Santiago, V., Ribeiro, A., Johnson Jr, R..L. et al. 2022. Modeling and Economic Analyses of Graded Particle Injections in Conjunction with Hydraulic Fracturing of Coal Seam Gas Reservoirs. *SPE Journal*: 1–16. <https://doi.org/10.2118/208404-PA>.
- Scott, M., Johnson, R.L., Jr., Datey, A. et al. 2010. Evaluating Hydraulic Fracture Geometry from Sonic Anisotropy and Radioactive Tracer Logs, SPE 133059. Proc., SPE Asia Pacific Oil and Gas Conference and Exhibition, Brisbane, Queensland, Australia, 18-20 October. <https://doi.org/10.2118/133059-MS>.
- Tiab, D and Donaldson, E.C. (2016): Petrophysics. – Gulf Professional Publishing, DOI: <https://doi.org/10.1016/C2014-0-03707-0>
- Undershultz, J. Mukherjee, S., Wolhuter, A., Xu, H., Banks, E., Noorduijn, S., and McCallum, J. (2019): Assessment of Fault Zone Properties for CSG Development Areas. – URTEC-198331-MS.
- Zhou, F., Oraby, M., Luft, Y., Guevara, M.O., Keogh, S. & Lai, W. (2022): Coal seam gas reservoir characterisation based on high-resolution image logs from vertical and horizontal wells: A case study. – *Int. J. of Coal Geology*, v. **262**, 1 October 2022, 104110 (<https://doi.org/10.1016/j.coal.2022.104110>).